

Cerner → Smile CDR

MoDHS interoperability layer - shared patient view

Date: 16th April 2026

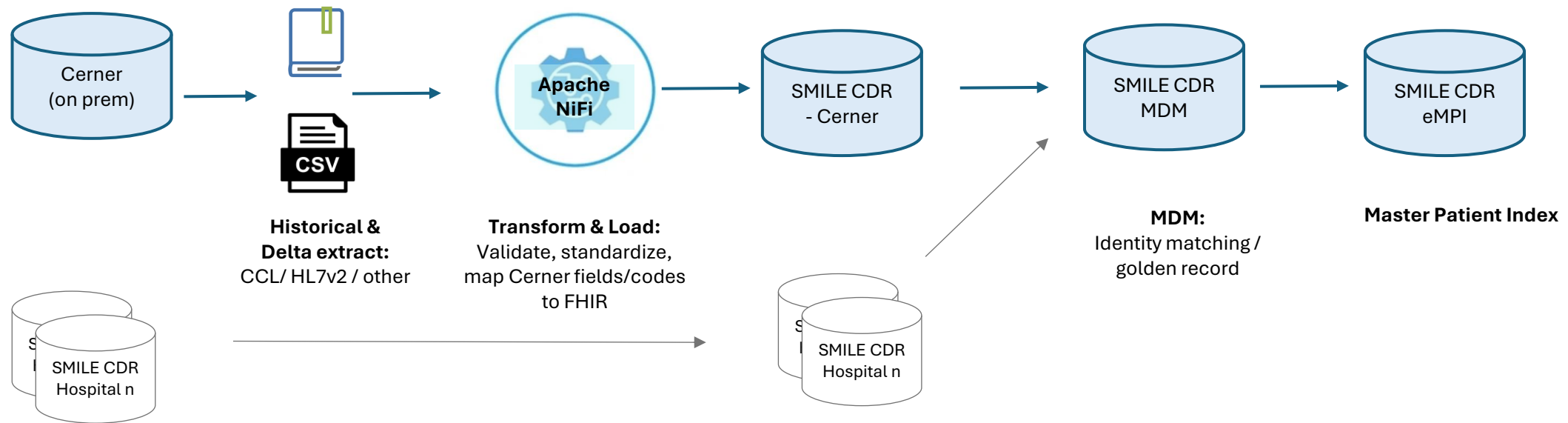
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Document Purpose:

- Bring all Cerner stakeholders to a common understanding of the MoDHS interoperability objective and the role of Cerner in that target architecture
- Align on the best supported approach for **historical** and **delta** data loads from Cerner into Smile CDR
- Review the first-pass Cerner-to-FHIR mapping and the supporting code set / identity questions
- Agree on cadence of working sessions needed to define extracts, mappings, and wider resource ingestion into Smile CDR
- This will remain a live / working document for engagement with Cerner

Cerner → Smile CDR: what we are doing now

- The target state is a unified FHIR-based data exchange platform using Smile CDR to standardise and create consolidated golden records across hospitals to linkage records across hospitals.
- The outcome to remove silo data, create a is a shared patient view.



Cerner is an important source feeding the MoDHS interoperability layer, alongside the other hospital EMRs.

FHIR Resources to be Populated with Cerner Data



MoDHS Solutions Architecture - Cerner Migration (1).pdf

1. Practice/ Hospital/Clinic	2. Core Resources & Appointment	3. Clinical facts and results	4. Medication fulfilment/resources	5. Documents, devices, and traceability
Organization	Person	Encounter	Medication	DocumentReference
Location	Patient	EpisodeOfCare	MedicationAdministration	Device
HealthcareService	RelatedPerson	CareTeam	MedicationDispense	ImagingStudy
	Practitioner	CarePlan	MedicationStatement	
	Consent	Goal	MedicationRequest	
	FamilyMemberHistory	Condition	NutritionOrder	
	Appointment	Observation	ServiceRequest	
	Schedule	Procedure	AllergyIntolerance	
	Slot	DiagnosticReport		
		Specimen		
		Immunization		

Cerner Engagement - Immediate next steps

We are engaging Cerner now because it is a significant on-premise source that must fit into the same standard ingestion and identity resolution model same as the other hospitals. We will break into 2 parts:

- initial historical bulk load,
- delta updates.

We need from Cerner

- Provide volumetrics
- Recommended supported mechanism for:
 - historical extract
 - delta/change extract
- Confirm whether existing outbound extracts or existing downstream CDR loads can be reused
- Start with working review of the initial Cerner-to-FHIR mapping – Patient
 - Confirm relevant patient/person/relationship code sets and how CODE_VALUE/CODE_SET should be interpreted
- Confirm authoritative source keys for identity:
 - National ID
 - Iqama
 - MRN
 - Person / patient identifiers used in Cerner

Cerner Engagement

What Cerner has shared so far

- Cerner Data Model
- A Cerner outbound integration specification for near real-time ADT messaging: the PDF explains Cerner Millennium's outbound HL7 ADT interface, including supported trigger events, segment structures, configurable outbound content, patient/encounter identifiers, and acknowledgment behaviour.
- A set of Cerner CCL extraction/query scripts for patient profile and clinical data: the text files provide runnable/queryable extracts for encounter, diagnosis, practitioner, prescription/pharmacy, lab, and radiology data, including paired jhp_patprofile_* and jhp_pp_get_* variants.

What the i-Secure/Smile team have done so far

- Mapped patient onboarding journey
- Requested data volumetrics
- Reviewed the data model and shared artefacts
- Created a first draft Cerner Patient/Person to FHIR Patient in readiness for a review with Cerner Team

Cerner – Discussion points to date

Discussion Points	ask	Updates/Notes
1. Existing Data Extracts and re-use	Are there existing integrations / extracts already in place – lets start with Patient demographic but also consider the other data in scope. How many can we re-use to for the data domains in scope? Who owns and supports these interface / integration? i.e. Is Patient Demographic data been loaded into the other CDR ((database). Can we re-use that extract?	15/04/2026: ADT outbound specifications Useful for understanding Cerner’s real-time patient and encounter event model, including person/patient updates, next-of-kin, allergies, diagnosis, guarantor and insurance segments. They do not yet provide the full extract design needed for a large historical migration Further discussion required
2. Constraints	Confirm any operational limits: batch windows, API limits, reporting impact, file size, bandwidth, and change control	
3. Data Structure Definition, CodeSystem, Conceptmaps	Review our draft Person/Patient mapping and confirm the correct source fields, keys, and business meaning referenced. Provide the local Cerner code sets used by the in-scope data, starting with Person/Patient based on the mapping document provided.	30/03/2026: Patient - Cerner to FHIR mapping (draft) document shared. We need to review with Cerner team. Meeting to be scheduled to review. 15/04/2026: Mapping document to be reshared with Cerner team
4. CodeSystem & Terminologies	Mapping documentation from Cerner local codes to standards already used by Cerner e.g. SNOMED CT, LOINC, ICD-10, CPT, HL7 administrative codes	
5. Data Volumetric	Initial volumetrics requested from cerner team: Number of patient records, active vs total Expected changes per day to patient record per day and week Number of forward looking appointment Expected number of changes to appointments records per day/week. Please refer to discovery document	15/04/2026: Ashjan provided Cerner team with approvals to share details.

Cerner → Smile CDR: what we are doing now

- We are starting with **Patient / Person / RelatedPerson** because identity and family linkage are foundational
- We already have a **first draft Cerner-to-FHIR mapping** for patient data to drive the discussion

#	FHIR Path	Element Type	Cardinality	Mapping Type	Source Field
0.1 Resource Qualification & Metadata					
0.1.1	RelatedPerson (resource instantiation rule)	rule	1..1	Conditional	PERSON_PERSON_RELTN.PERSON_PERSON_RELTN_ID
0.1.2	RelatedPerson.meta.lastUpdated	instant	0..1	Transform	PERSON_PERSON_RELTN.UPDT_DT_TM; related PERSON.UPDT_DT_TM; related PERSON_NAME.UPDT_DT_TM; related PERSON_ALIAS.UPDT_DT_TM
0.1.3	RelatedPerson.meta.source	uri	0..1	Fixed	-
0.1.4	RelatedPerson.active	boolean	0..1	Transform	PERSON_PERSON_RELTN.ACTIVE_IND; PERSON_PERSON_RELTN.ACTIVE_STATUS_CD; PERSON_PERSON_RELTN.END_EFFECTIVE_DT_TM
1.1 Relationship and source identifiers					
1.1.1	RelatedPerson.identifier[source-relationship-id].value	string	1..1	Direct	PERSON_PERSON_RELTN.PERSON_PERSON_RELTN_ID
1.1.2	RelatedPerson.identifier[source-relationship-id].system	uri	1..1	Fixed	-
1.1.3	RelatedPerson.identifier[source-related-person-id].value	string	0..1	Direct	PERSON_PERSON_RELTN.RELATED_PERSON_ID
1.1.4	RelatedPerson.identifier[source-related-person-id].system	uri	0..1	Fixed	-
1.1.5	RelatedPerson.identifier[MRN].value	string	0..*	Transform	related PERSON_ALIAS.ALIAS; related PERSON_ALIAS.PERSON_ALIAS_TYPE_CD; related PERSON_ALIAS.ALIAS_POOL_CD
1.1.6	RelatedPerson.identifier[national-id].value	string	0..*	Transform	related PERSON_ALIAS.ALIAS; related PERSON_ALIAS.PERSON_ALIAS_TYPE_CD; related PERSON_ALIAS.ALIAS_POOL_CD
1.2 Link to patient and relationship period					
1.2.1	RelatedPerson.patient.identifier.value	string	1..1	Direct	PERSON_PERSON_RELTN.PERSON_ID
1.2.2	RelatedPerson.patient.identifier.system	uri	1..1	Fixed	-
1.2.3	RelatedPerson.patient.reference	string	0..1	Transform	PERSON_PERSON_RELTN.PERSON_ID
1.2.4	RelatedPerson.period.start	dateTime	0..1	Direct	PERSON_PERSON_RELTN.BEG_EFFECTIVE_DT_TM
1.2.5	RelatedPerson.period.end	dateTime	0..1	Direct	PERSON_PERSON_RELTN.END_EFFECTIVE_DT_TM
2.1 Relationship coding and local flags					
2.1.1	RelatedPerson.relationship[0].coding[0].code	code	1..*	Transform	PERSON_PERSON_RELTN.FAMILY_RELTN_SUB_TYPE_CD; PERSON_PERSON_RELTN.PERSON_RELTN_CD; PERSON_PERSON_RELTN.CONTACT_ROLE_CD
2.1.2	RelatedPerson.relationship[0].text	string	0..1	Transform	PERSON_PERSON_RELTN.FAMILY_RELTN_SUB_TYPE_CD; PERSON_PERSON_RELTN.PERSON_RELTN_CD; PERSON_PERSON_RELTN.CONTACT_ROLE_CD

Reviewing the Cerner to Patient Mapping document

CODE SETS & THE CODE_VALUE TABLE

We understand all coded fields resolve to a single CODE_VALUE table (including CODE_SET). We would like to:

- Confirm how to look up valid values for a given code set (e.g. Code Set 351 for relationship types)
- Understand whether code sets are consistent across all MoDHS Cerner hospitals, or configured per the six hospitals in scope:
 - Prince Sultan Military Medical City (Riyadh)
 - Prince Sultan Cardiac Centre (Riyadh)
 - Al Kharj Military Industries Corporation Hospital (Al Kharj)
 - Armed Forces Hospital (Al Kharj)
 - Prince Sultan Airbase Hospital (Al Kharj)
 - Armed Forces Hospital (Qaseem)

Confirmed Cerner is single tenant setup – single set of valuesets / codesets

FHIR RESOURCES mapping - Start small with Patient

With reference to the attached PowerPoint (for near term focus)

- **Person/Patient** - PERSON_PATIENT
 - Key fields and identifiers
 - Validate our mapping that we will share in meeting

We have provided a first draft Mapping document for Patient.

TASK: review initial draft mapping of Cerner to FHIR Person/Patient.

1. Secure understanding of which fields we should be sourcing following our draft mapping from Cerner to Person/Patient.
2. Get tract of related codesets (as per code set list in the mapping spreadsheet) for Person/Patient
3. Discussion – Is Patient Demographic data been loaded into the other CDR (database). Can we re-use existing that extract ?

The first iteration review of mapping document will help us understand if we have mapped correctly, and then we can refine mapping document to drive extract format.

Engagement Cadence

Initial cadence

- Start with **2 working sessions per week**
- Use this initial cadence for the next **2–4 weeks**
- Then review and move to a sustainable rhythm based on progress and dependencies

Focus of the first phase

- Close immediate open questions on:
 - Patient / Person / RelatedPerson
 - identity keys
 - CODE_VALUE / CODE_SET
 - volumetrics
 - bulk historical extract approach
 - delta / change synchronisation approach
 - define **master/reference extracts** for:
 - Organization , Location , HealthcareService , Practitioner , scheduling / service reference data
- Progress through additional resources in a controlled sequence:

Engagement Cadence Calendar

[illegible]

Cerner –Proposed Approach:

Historical Bulk Load

- **Goal:** one-time extraction of full history for patients, encounters, and clinical data
- **Approach:** Cerner-owned CCL/file bulk extracts and leverage existing extract code/mechanisms where possible
- **Why:** better suited than HL7v2 event messages for large historical volumes
- **Needs:** extract scope, file layout, code sets, master data, record counts.

Deltas Loads

- **Goal:** capture new and changed records after initial bulk load, leverage existing mechanisms/code
- **Approach:** use HL7v2 feeds where available, plus delta extracts where needed
- **Scope:** patient/admin events, encounter changes, orders, results
- **Needs:** watermark logic, trigger coverage, replay/recovery rules.

Supplemental Delta Extracts

- **Goal:** keep non-message-supported data domains in sync
- **Approach:** scheduled Cerner-owned delta extracts
- **Scope:** reference/master data and domains not reliably covered by HL7v2
- **Needs:** update timestamp field, delete/merge handling, refresh frequency.

Cerner – 41 resources to be loaded



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		Specimen		
		Immunization		

Cerner – FHIR resources to be loaded

Groups	FHIR Resource	FHIR documentation	Bulk Extract Mechanism	Delta updates Mechanism	What Cerner have provided so far
Reference / Foundations	Organization	https://hl7.org/fhir/R5/organization.html https://portal.nphies.sa/ig/StructureDefinition-organization.html	Cerner master file CCL extract		
	Location	https://hl7.org/fhir/R5/location.html https://portal.nphies.sa/ig/StructureDefinition-location.html	Cerner master file CCL extract		
	HealthcareService	https://hl7.org/fhir/R5/healthcareservice.html	Cerner master file CCL extract		
Core Resources & Appointment	Person	https://hl7.org/fhir/R5/person.html	CCL script?		
	Patient	https://hl7.org/fhir/R5/patient.html https://portal.nphies.sa/ig/StructureDefinition-patient.html	CCL script?		
	RelatedPerson	https://hl7.org/fhir/R5/relatedperson.html			
	Practitioner	https://hl7.org/fhir/R5/practitioner.html			
	Consent	https://hl7.org/fhir/R5/consent.html			
	FamilyMemberHistory	https://hl7.org/fhir/R5/familymemberhistory.html			
	Appointment	https://hl7.org/fhir/R5/appointment.html			
	Schedule	https://hl7.org/fhir/R5/schedule.html			
	Slot	https://hl7.org/fhir/R5/slot.html			

Cerner – FHIR resources to be loaded

Groups	FHIR Resource	FHIR documentation	Bulk Extract Mechanism	Delta updates Mechanism	What Cerner have provided so far
Clinical & Orders and requests	Encounter				
	EpisodeOfCare				
	CareTeam				
	CarePlan				
	Goal				
	Condition				
	Observation				
	Procedure				
	DiagnosticReport				
	Specimen				
	Immunization				
	Encounter				
	EpisodeOfCare				
	CareTeam				

Groups	FHIR Resource	FHIR documentation	Delta Extract Mechanism	Delta Extract Mechanism	What Cerner have provided so far
Medication fulfilment	Medication				
	MedicationAdministration				
	MedicationDispense				
	MedicationStatement				
Documents, devices, and traceability	DocumentReference				
	Device				
	Provenance				

Plan/ Timelines

- Completed mapping for Bulk by end of June 2026
- Day +1 deltas - TBC

Cerner – Discussion points from previous meetings

NATIONAL ID & IQAMA VALIDATION — CRS & INTEGRATION

We would like to understand how Cerner uses the CRS (Civil Registration System) and the National Health Core Registry as controls/ validation to validate National IDs and Iqama numbers at the point of Patient registration or in other processes.

We need confirmation if Cerner use the National Health Core Registry which means would confirm controls around validity of National ID/Iqama and demographic data

VOLUMETRICS — DISCOVERY SHEET

We would like to complete the volumetrics discovery sheet for Cerner.

- Number of patient records, active vs total
- Expected changes per day to patient record per day and week
- Number of forward looking appointment
- Expected number of changes to appointments records per day/week. Please refer to discovery document

Discovery Sheet to be sent by Lili-Ann Gustafsson lili@isecure-solutions.com – permissions from MoDHS in place for Cerner team to provide initial volumetric information

Integration with Cerner

1. What is Cerner's recommended supported pattern for batch and delta outbound integration to an external FHIR repository like Smile CDR?
2. What identity keys and event patterns should we use so Smile MDM/EMPI can match, update, and de-duplicate safely?
National ID, Patient ID MRN etc

Next step Extended sessions to be indicated to review mapping doc -

Cerner – FHIR resources to be loaded - High level list

Person	RelatedPerson	Patient	CodeSystem
Consent	FamilyMemberHistory	Practitioner	ConceptMap
PractitionerRole	Organisation	Location	StructureDefinition
Appointment	Slot	Schedule	NamingSystem
HeathcareService	Medication	Procedure	Provenance
Observation	Encounter	Device	DocumentReference
AllergyIntolerance	CarePlan	CareTeam	ServiceRequest
EpisodeOfCare	Condition	DiagnosticReport	Order
Goal	Immunization	MedicationDispense	
MedicationRequest	MedicationAdministration	MedicationStatement	
ImagingStudy	Specimen	NutritionOrder	

File Server Export & Cleaning, Cerner Migration
Version 1.0 - 09 December 2020

